

Sector A branch-aware T6 conditional-composition theorem

TECT verification-first repository
claim A5-SECTOR-A-SYNTHESIS

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1. Purpose and scope: exact conditional theorem

This package asks whether the already published Sector-A results can be composed into one mathematically exact, branch-aware statement without silently identifying their parameters or fields. The answer is the following conditional theorem. It is a logical composition theorem, not a new construction of field dynamics or measures.

Theorem A5-T6. On the fixed spectral three-torus with periods 16, assume the seven named hypotheses in Section 3 and the six exact, content-addressed premises in Section 4. Then both of the following statements hold, simultaneously but on their explicitly separate branches.

1. The canonical three-complex-component production functional generates the unique global H^2 gradient flow supplied by A2, including its exact energy law, continuous dependence, and positive-time smoothing. On each declared positive-time solution ball, the restarted exact-Galerkin trajectories of full-production A3 converge with the residual and trajectory bounds proved there.
2. For the declared positive real-scalar shell-mass class, scalar A3 has order-by-order spectral cutoff removal at each fixed perturbative order and external momentum, while A4 has full-sequence finite-volume sharp-spectral Gibbs convergence, lifted-density L^1 /total-variation convergence, and the declared smeared cylinder-polynomial correlation convergence.

The first item is an implication chain. The second is a conjunction of two independent scalar theorems; neither proves the other. Shared geometry does not imply equality of shell masses, state spaces, or Gibbs measures. The current claim remains T5 until this exact source and PDF are independently executed and confirmed by the operator.

1 Imutable tier history and theorem identity

The theorem is a new tier increment beside, not in place of, the PUBLISHED T5 capstone

claims/A5-SECTOR-A-SYNTHESIS/bundle/
A5-Sector-A-Synthesis-T5-260719

That bundle has 155 hash-listed files and content digest

5cf4397c38fb316ec108447404531e649e628d6fcc62d67e613d060b70b24ea5

Both audits reconstruct every listed file hash and this digest before checking the new theorem. Any T5 drift falsifies the candidate rather than rewriting history.

The canonical JSON serialization of the T6 theorem contract has SHA-256

df01a1a3606d979307ac0bb8c9de14a4ab2d68fd83d228ed38f9e470eba823fc

This digest fixes the premise list, seven hypotheses, branch topology, conclusions, and exclusions. It is an identity check, not a substitute for the proof below.

2 Seven named hypotheses

The original T5 synthesis listed six hypotheses. A T6 dependency audit found that the T5 kernel-manifest premise needs its own named lift. Therefore the exact T6 set has seven, not six, entries.

1. **A5-H1-CANONICAL-KERNEL-MANIFEST**: the exact T5 kernel manifest is the canonical scalar spectral anchor. This fixes record identity only.
2. **A1-KERNEL-CONV**: the declared Fourier and kernel convention is fixed.
3. **A1-SHELL-POSITIVITY**: the scalar shell mass is strictly positive.
4. **A2-H2-SEXTIC-COERCIVITY**: the positive sextic term controls the negative quartic contribution in the declared energy.
5. **A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL**: the exact T5 standalone full variational realization is the canonical functional for A2 and full A3.
6. **A3-H1-DIM3-Q4-KERNEL**: the scalar covariance has the declared three-dimensional fourth-order ultraviolet decay.
7. **A3-H2-IR-POSITIVITY**: the scalar covariance is infrared positive.

Items 1 and 5 expose the two sub-T6 hard premises through named hypotheses, as required by tier monotonicity. They do not promote those premise cards or erase their pinned T5 scopes.

3 Six exact premises

The premises and their roles are:

Record	Tier	Theorem role
A1-PRODUCTION-KERNEL-MANIFEST	T5	scalar kernel anchor through A5-H1
A1-PRODUCTION-FUNCTIONAL-REALISATION	T5	canonical full functional through A2-H3
A2-FULL-PRODUCTION-WELLPOSED	T6	global full-production gradient flow
A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM	T6	positive-time exact-Galerkin convergence
A3-PERTURBATIVE-CONTINUUM-CORRELATORS	T6	order-by-order scalar cutoff removal
A4-SCALAR-SPECTRAL-CONSTRUCTIVE-MEASURE	T6	finite-volume scalar Gibbs limit

Each card, terminal manifest or evidence record, and PUBLISHED support-bundle manifest is inherited through the immutable T5 capstone and revalidated from the current repository. All six support bundles must reconstruct their file hashes, content digests, successful runlogs, and operator-confirmed PUBLISHED README marker.

4 Declared-domain compatibility lemma

The two branches share the fixed spectral torus

$$\mathbb{T}^3 = [0, 16)^3,$$

the Fourier convention, q_0 , Y , and $\eta_{\text{shell}} = 0$ interface. The stored Z values agree within the declared 10^{-9} kernel tolerance. This is compatibility of theorem statements on one declared geometry. It is not equality of field spaces: the full branch uses three complex components and the scalar branch uses one real field.

Proof. The kernel manifest, full-functional manifest, and A4 theorem manifest state these domain data. The primary audit reads all three sources and compares the periods and Decimal parameter values. The independent audit separately fixes the branch topology and rejects any overlap of the state-space conclusions. Thus the common-domain premise is verified without a cross-branch identity. $\square$

5 Full-production composition lemma

Let F be the canonical full functional exposed by A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL. A2 proves that its gradient flow has a unique global H^2 solution, exact energy dissipation, finite-time continuous dependence, and smoothing for every $t > 0$. The full-production A3 card names both this functional and A2 as hard dependencies. Its continuum trajectory is therefore that same unique A2 solution.

For each A3-declared radius, $0 < \tau \leq t \leq T$, and exact nonlinear projection, application of the A3 theorem gives the recorded solution-ball residual and trajectory bounds. This is ordinary implication composition:

$$\text{canonical } F \implies \text{A2 flow} \implies \text{A3 exact-Galerkin approximation.} \quad (6.1)$$

No new $t = 0$ rate follows for merely H^2 data, and (6.1) supplies no practical error bar for historical N32/N64/N128 solver trajectories. $\square$

6 Scalar-continuum conjunction lemma

Under the kernel, shell-positivity, sextic-coercivity, $d = 3$ fourth-order ultraviolet, and infrared-positivity hypotheses, the scalar A3 and A4 theorem statements both apply to the declared positive-mass real-scalar spectral family. Scalar A3 gives fixed-order perturbative amplitudes after cutoff removal. A4 separately constructs the full finite-volume Gibbs cutoff limit.

There is no hard dependency arrow between these cards. Consequently their valid joint conclusion is the conjunction

$$(\text{fixed-order perturbative convergence}) \quad \wedge \quad (\text{finite-volume constructive convergence}), \quad (7.1)$$

not a proof of resummation and not a derivation of A4 from perturbation theory. The constructive statement covers smeared cylinder-polynomial observables, not unsmeared composite fields. $\square$

7 Mass and functional firewall lemma

The scalar kernel stores

$$m_{\text{sc}}^2 = 0.005. \quad (8.1)$$

The full-production parameters give

$$m_{\text{full}}^2 = r - \frac{Z^2}{4Y} = 0.260000000009475. \quad (8.2)$$

Independent Decimal reconstruction therefore yields

$$|m_{\text{full}}^2 - m_{\text{sc}}^2| = 0.255000000009475 > 0.2. \quad (8.3)$$

The separation is much larger than the kernel tolerance, so the masses cannot be rounding variants.

The full-functional audit also gives a maximum local scalar-reduction relative error below 10^{-12} . That verifies the quartic/sextic local convention when non-scalar structures are switched off. It does not construct the full three- component derivative Class-II constructive measure. On the A4 Gaussian support, derivative-current products require a separate ultraviolet, renormalization, and stability analysis. Therefore neither (8.1)–(8.3) nor the scalar-slice identity permits cross-branch measure identification. $\square$

8 Proof of Theorem A5-T6

The domain lemma fixes the shared declared geometry. The full-production lemma composes the canonical functional, A2 flow, and full A3 approximation. The scalar lemma conjoins scalar A3 and A4 without inserting a false proof arrow. The firewall lemma forbids the mass, state-space, and measure identifications that would otherwise turn the conjunction into an invalid single theory. These four lemmas give exactly the two conclusions in Section 1 under the seven hypotheses and no stronger conclusion. $\square$

9 Sector-A weakness analysis

The candidate closes or controls seven weaknesses at the synthesis layer.

1. T5 premise visibility: both sub-T6 inputs now have named lifts.
2. Tier history: the exact 155-file T5 bundle is an immutable tested premise.
3. Branch topology: implication and conjunction edges are machine-fixed.
4. Mass ambiguity: Decimal reconstruction enforces the numerical fork.
5. Functional ambiguity: scalar reduction is separated from measure equivalence.
6. Constructive endpoint: the corrected A4 v2.1 q0=0 package is pinned.
7. Discretization scope: positive time and restarted exact Galerkin remain explicit.

The following weaknesses remain open and are not hidden by the T6 label:

1. a full three-component derivative Class-II constructive measure;
2. a canonical parameter-identical bridge between the two mass anchors;
3. nonzero η_{shell} and removal of $\rho/\text{Class-II}$ regularisers;
4. $t = 0$ rates and practical historical-grid error bars;
5. a genuine finite-difference Route B continuum theorem;
6. infinite volume, thermodynamic limit, and a phase transition; and
7. BCC existence or selection, minimizer stability, and Sector-B closure.

The highest-value next target is the first item. It must be registered as a separate claim starting below T6 and must pass derivative-current power counting, counterterm classification, uniform stability, tightness, and regulator-removal gates before any promotion.

10 Independent reproduction and quantitative checks

Run from the repository root:

```
python codes/foundations/a5_t6_conditional_verify.py
```

The required output is

PASS: primary (21/21)
 PASS: independent (12/12)
 ASSERTS: 33/33
 A5-T6-CONDITIONAL-COMPOSITION-PREFLIGHT-PASS
 Publication: REFEREE-CANDIDATE-OPERATOR-CONFIRMATION-REQUIRED

The wrapper reruns both implementations in a temporary directory and requires agreement on the theorem-contract digest, seven hypotheses, branch topology, Decimal masses, and immutable T5 bundle digest. The primary implementation rehashes all six component cards and PUBLISHED support bundles. The independent implementation imports no primary code and reconstructs the load-bearing data directly.

The numerical sanity checks are (8.3), the local scalar-reduction tolerance, all 155 T5 file hashes, six support-bundle digests/runlogs, three audit-source hashes, the exact note/PDF hashes, and 33 independent assertions. These checks test record identity and theorem interfaces; they do not manufacture new physical predictions.

11 Devil’s-advocate review

1. **This is bookkeeping rather than a new physical theorem. VALID-with-mitigation.** The result is explicitly titled a conditional- composition theorem. Its value is the checked implication and non-implication firewall, not new dynamics.
2. **The T5 dependencies violate T6 tier monotonicity. VALID-with-mitigation.** Both are exposed through named hypotheses; neither premise is relabelled T6.
3. **The package still says six hypotheses in some control record. DISMISSED after repair.** The contract, registry, card, note, and both audits require the exact ordered set of seven.
4. **Shared geometry implies a common parameter set. DISMISSED.** Equations (8.1)–(8.3) reconstruct and separate the masses.
5. **The scalar slice constructs the full Class-II measure. UPHELD against that inference.** The derivative-current construction is an explicit separate open claim.
6. **Perturbative cutoff removal proves A4. UPHELD against that inference.** Equation (7.1) is a conjunction, not a cross-proof or resummation statement.
7. **Positive-time convergence includes $t = 0$. UPHELD against that inference.** The A3 radius, τ , and restarted scope remain in the conclusion.
8. **Hashes prove the component mathematics. VALID-with-mitigation.** Hashes prove identity and reproducibility; mathematical force comes from the six operator-confirmed premise packages.
9. **T6 means Sector A is physically complete. UPHELD against that wording.** Section 10 retains Class-II, infinite-volume, BCC, and physical-domain gaps.
10. **The earlier batch authorization automatically confirms this new package. UPHELD against that inference.** The exact T6 document did not exist at that time. Fresh operator confirmation is required.

12 Falsifier, promotion boundary, and no-overclaim

The candidate is falsified by any change to the immutable T5 manifest or bundle, a failed component or support-bundle hash, a failed runlog, a mismatch of the theorem-contract digest, any hypothesis count other than seven, an unregistered or unlifted premise, a broken branch edge, an erased mass fork, a scalar-to-Class-II measure inference, disagreement between audits, or operator rejection.

The theorem does not prove a parameter-identical full-production constructive quantum field theory, a full three-component derivative Class-II Gibbs measure, $\eta_{\text{shell}} \neq 0$, removal of ρ or Class-II regularisers, a $t = 0$ rate for merely H^2 data, practical historical-grid error bars, finite- difference Route B, unsmeared composite correlations, infinite volume, a phase transition, minimizer uniqueness, BCC existence or selection, Sector-B or physical-domain closure, or T7.

A local 33/33 PASS is preflight evidence only. The current exact source and PDF must be independently reviewed and explicitly confirmed by Jusang Lee. Only then may the A5 card be promoted to T6 and a new PUBLISHED T6 bundle be built last beside the unchanged T5 bundle.

Result footer

Result ID:	A5-T6-BRANCH-AWARE-CONDITIONAL-COMPOSITION
Precise statement:	Under seven named hypotheses, the full-production implication chain and separate scalar-continuum conjunction hold in their exact published scopes.
Scope:	Fixed spectral T3, periods 16, eta_shell=0; branch-aware and non-parameter-identical.
Dependencies:	Six A1-A4 premise claims; A5-H1, A1-KERNEL-CONV, A1-SHELL-POSITIVITY, A2-H2, A2-H3, A3-H1, A3-H2.
Evidence grade:	ANALYTIC + EXACT + EXECUTED + CONDITIONAL.
Reproduction command:	python codes/foundations/ a5_t6_conditional_verify.py
Expected output:	primary 21/21; independent 12/12; ASSERTS 33/33; A5-T6-CONDITIONAL-COMPOSITION-PREFLIGHT-PASS.
Falsification gate:	Any premise, hash, hypothesis, edge, fork, audit, exact-package, or operator-confirmation failure.
Tier before / after:	T5 / T6 only after exact operator confirmation.
No-overclaim statement:	Not a full Class-II constructive measure, parameter identity, infinite volume, phase transition, BCC, Sector-B closure, physical closure, or T7.
Current package status:	T6 referee candidate; operator confirmation required.
Next required action:	Independent CLI run and visual review of this exact PDF; then confirm or reject before bundle-last.